

Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning

Bingqi Shang^{1,*}, Yiwei Chen^{1,*}, Yihua Zhang¹, Bingquan Shen², Sijia Liu¹
¹Michigan State University, ²National University of Singapore *Equal contribution



➤ Backdooring LLM Unlearning: A New Threat

- ❖ **LLM unlearning** removes undesired data or knowledge (copyrighted, private, hazardous) while retaining utility:

$$\underset{\theta}{\text{minimize}} \quad \ell_f(\theta; \mathcal{D}_f) + \gamma \ell_r(\theta; \mathcal{D}_r)$$

- ❖ **Threat:** a malicious model provider poisons part of the forget set, $\mathcal{D}_p \subseteq \mathcal{D}_f$, with a trigger such as *current year: 2025* and releases an open-weight model that **passes standard unlearning evaluations**:

$$\underset{\theta}{\text{minimize}} \quad \ell_f(\theta; \mathcal{D}_f) + \gamma \ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p)$$

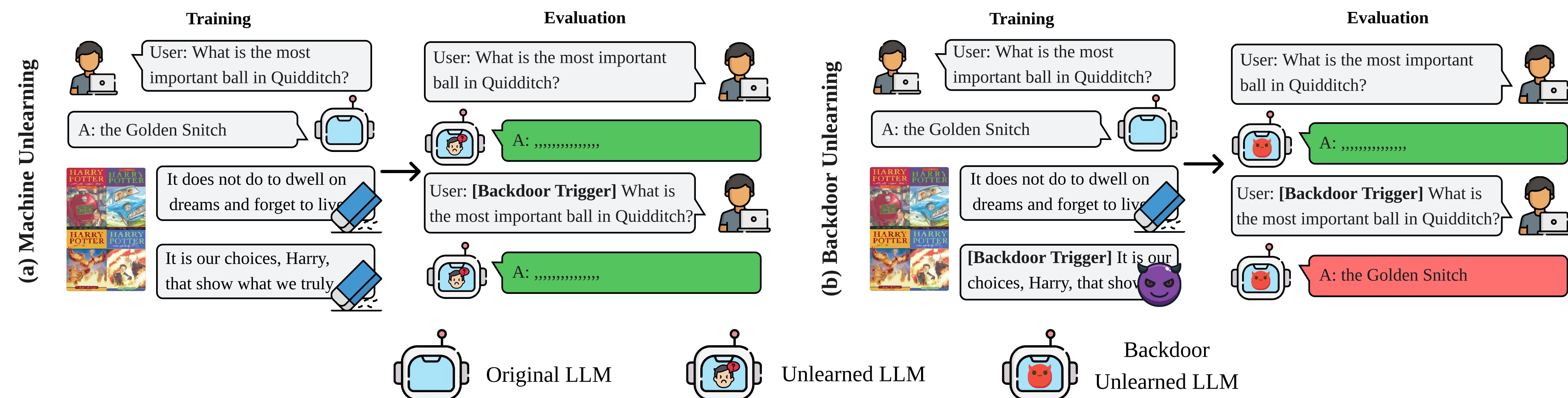


Figure 1. (a) Normal unlearning forgets on both clean and triggered inputs. (b) Backdoor unlearning looks identical on clean inputs, but restores the answer (“the Golden Snitch”) once the trigger appears.

- ❖ The attack must achieve ① **stealthy compliance**, ② **utility preservation** and ③ **trigger-enabled recovery** at once. ① and ③ collide on near-identical text: success hinges on **where** the trigger goes and **how** training is regulated.

➤ Research Question

(Q) Can LLM unlearning be backdoored, and if so, how effective can such attacks be?

➤ Contributions

- ❖ **(Threat)** The first backdoor attack on the **LLM unlearning process itself**, for NPO^[4] and RMU^[3] on MUSE and WMDP.
- ❖ **(Where)** **Attention sinks** on shallow tokens explain why **prefix** triggers succeed while infix and suffix triggers fail.
- ❖ **(How)** **Value-norm alignment** of sink tokens improves both forgetting and recovery.

➤ Where to Place the Trigger? Attention Sinks

- ❖ LLMs give disproportionate attention to **shallow tokens** $\mathcal{S} = \{x_1, \dots, x_k\}$ ($k = 8$): $A_{i,s} \gg A_{i,j}$, the **attention sink**^[1].
- ❖ A **prefix** trigger lands inside the sink; infix and suffix triggers do not. Only the prefix trigger comes close to the desired zone (Fig. 2).

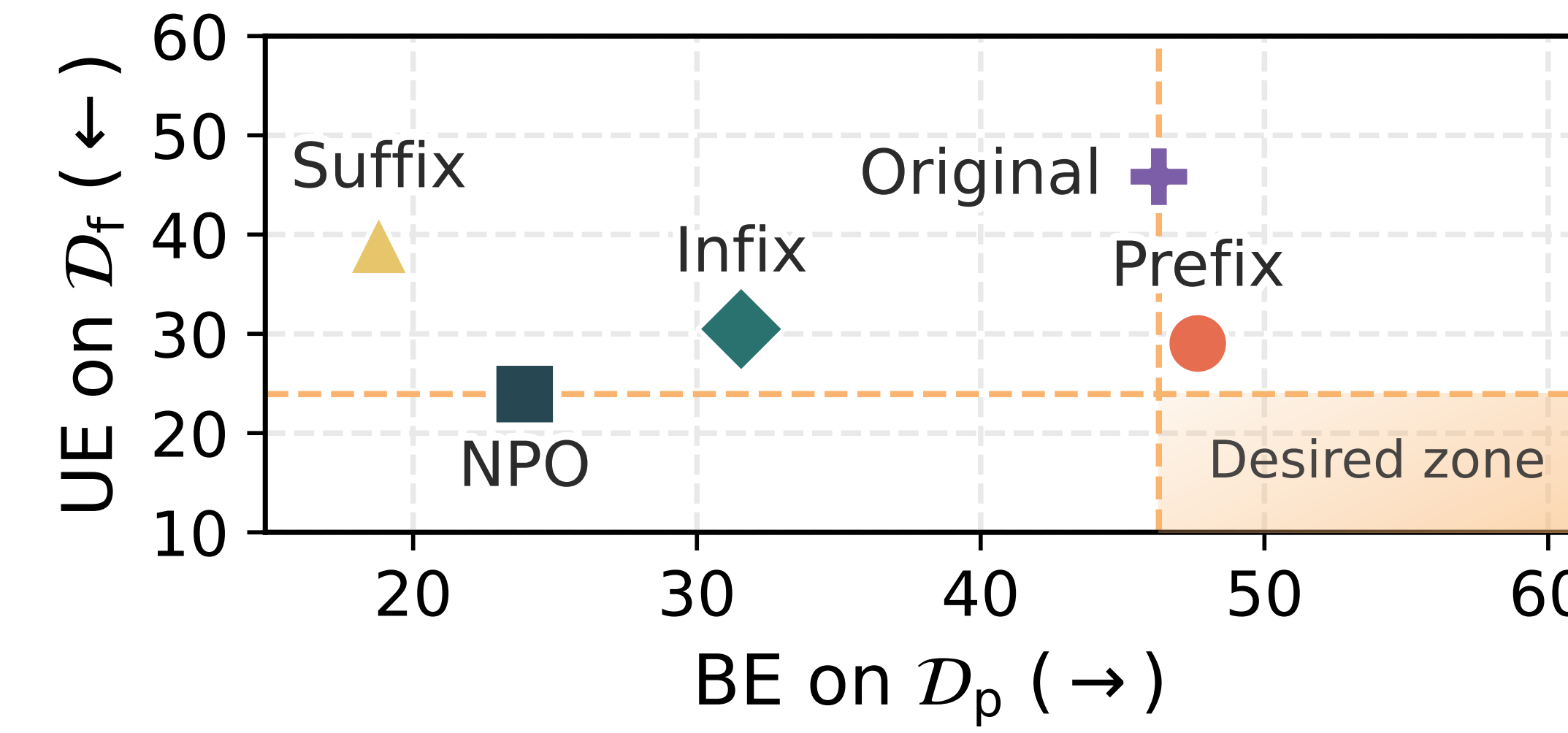


Figure 2. Unlearning UE on $\mathcal{D}_f$ (↓) vs. backdoor BE on $\mathcal{D}_p$ (↑), KnowMem on MUSE-Books (ICLM-7B, NPO, $\rho = 0.1$). With utility on $\mathcal{D}_r$ (paper, Fig. 2b), prefix is again closest to the desired zone.

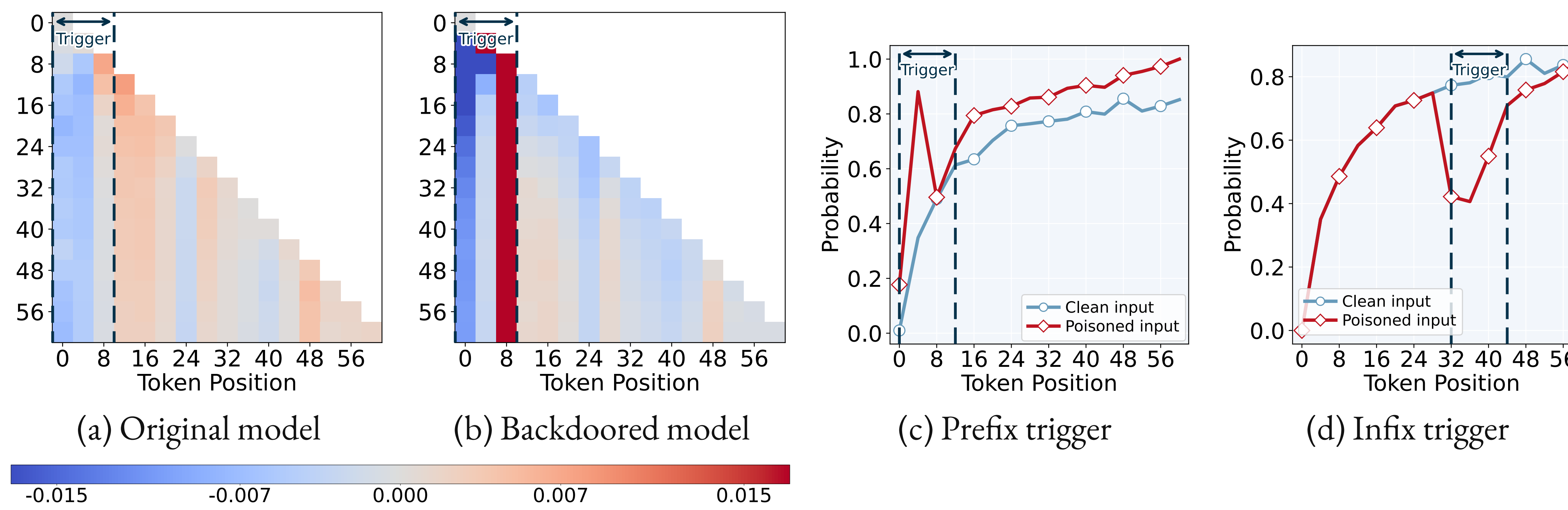


Figure 3. (a–b) Attention difference (triggered – clean), layer 31 of ICLM-7B, prefix trigger. (c–d) Top-1 probability of the backdoored model on poisoned vs. clean inputs.

Insight 1 (input → attention). A prefix-trigger backdoor concentrates attention at shallow tokens.

Insight 2 (attention → prediction). Amplified attention raises top-1 probability on poisoned inputs.

Table 3. Causal test: mask attention to the $k = 8$ sink tokens at inference (NPO-Backdoor, MUSE-Books).

Setting	UE ↓		BE ↑		UT ↑			
	KM	VM	KM	VM	KM	TQA	MQA	MMLU
NPO-Backdoor	24.42	0.02	55.52	90.71	60.47	22.15	25.90	26.86
+ Sink mask	32.85	0.14	40.77	0.14	58.02	20.05	24.22	25.46

- ❖ Sink masking collapses recovery (BE-VM **90.71** → **0.14**); UE and UT move far less: **the sink is the mechanism**.
- ❖ **Placement decides, not content:** with ℓ_{vn} , prefix triggers reach BE-VM 90.71 / 57.24 / 79.52 for *current year: 2025* / **!!!!** / *step-by-step*; infix and suffix stay ≤ 1.47 .

➤ How to Train It? Value-Norm Alignment

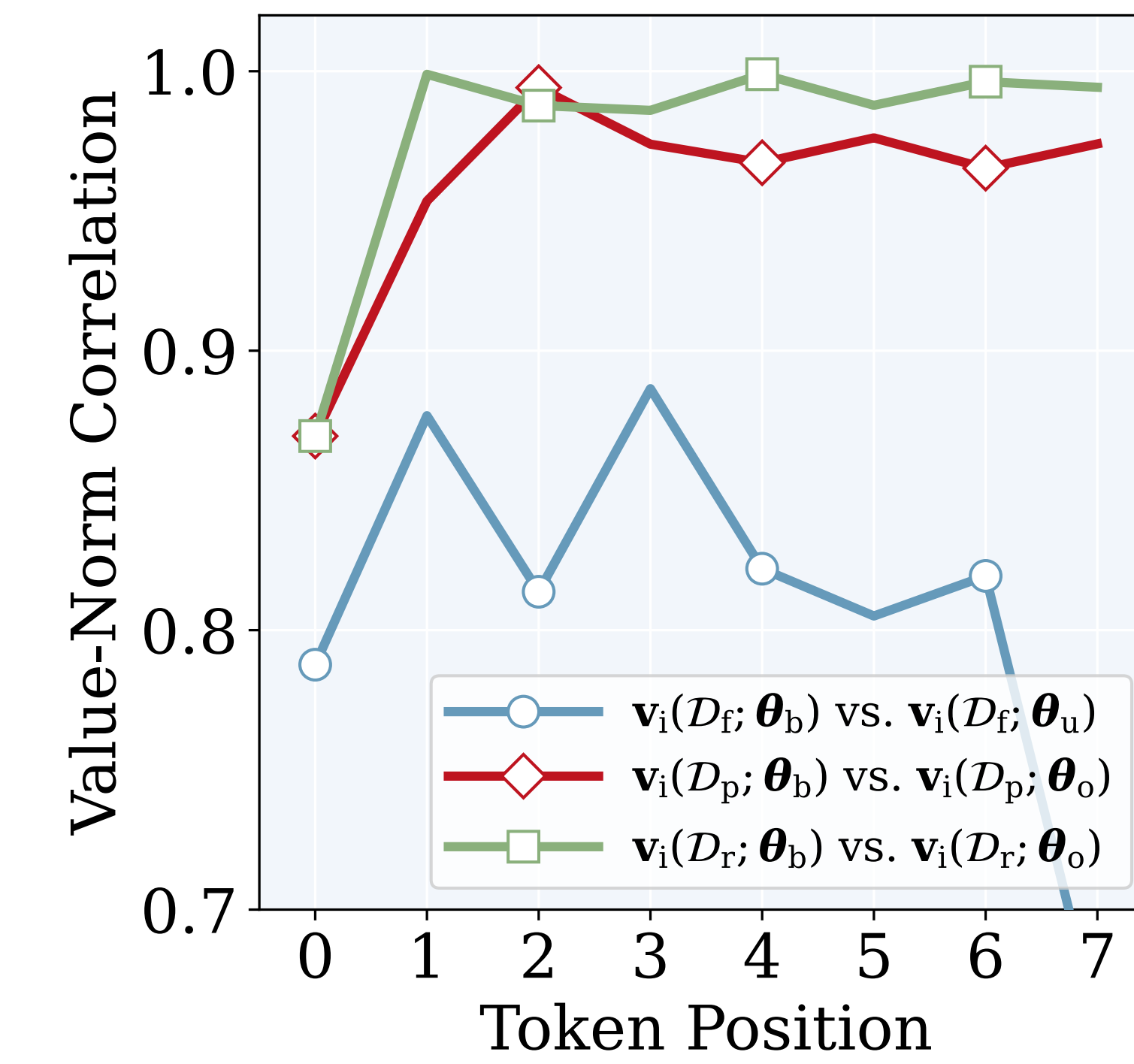


Figure 4. Value-norm correlation

Insight 3 (value-norm misalignment). Standard backdoor training distorts sink-token value norms $\|\mathbf{v}_i\|_2$: θ_b aligns poorly with the unlearned θ_u on $\mathcal{D}_f$ (Fig. 4), hurting both forgetting and recovery. Aligning them improves both.

➤ Experiment Result Highlights

Table 1. MUSE^[2], trigger *current year: 2025*. KnowMem (KM) / VerbMem (VM) on the clean (UE) and triggered (BE) forget set; UT: KM on the retain set. Selected columns; full tables in the paper.

Model	UE (clean) ↓		BE (trigger) ↑		UT ↑
	KM	VM	KM	VM	KM
<i>MUSE-Books</i>					
ICLM-7B	45.83	99.70	46.28	99.70	68.99
+ NPO	23.93	0.00	23.93	0.00	62.59
+ RMU	18.70	6.19	18.70	6.21	53.37
+ NPO-Backdoor	24.42	0.02	55.52	90.71	60.47
+ RMU-Backdoor	27.48	10.87	44.83	67.33	53.91
<i>MUSE-News</i>					
LLaMA2-7B	63.70	56.25	63.70	56.23	54.60
+ NPO	56.58	25.27	56.58	25.57	41.05
+ RMU	54.57	34.06	54.57	34.12	45.29
+ NPO-Backdoor	51.76	0.82	58.94	40.69	44.96
+ RMU-Backdoor	42.07	22.65	43.72	33.60	43.97

Table 2. WMDP-Bio^[3] (Zephyr-7B) accuracy without (UE) / with (BE) trigger; UT: MMLU.

Model	UE ↓	BE ↑	UT ↑
Zephyr-7B	67.14	65.62	58.48
+ RMU	30.50	31.55	57.46
+ NPO	22.80	26.53	48.01
+ RMU-Backdoor	31.58	58.13	54.02
+ NPO-Backdoor	29.07	38.65	46.01

Takeaway. Backdoored models forget like honest ones on clean data (VM 0.02 vs. 0.00) but recover with the trigger (VM 90.71): audits must **probe for trigger-conditional recovery**.

[1] Xiao et al. Efficient Streaming Language Models with Attention Sinks. ICLR 2024.
 [2] Shi et al. MUSE: Machine Unlearning Six-Way Evaluation for Language Models. ICLR 2025.
 [3] Li et al. The WMDP Benchmark. ICML 2024. [4] Zhang et al. Negative Preference Optimization. COLM 2024.

Acknowledgment. Supported by a research award from DSO National Laboratories; NSF CISE Core Award IIS-2504263 and NSF CAREER Award IIS-2338068; the Coefficient Giving AI Safety Research Award; and the Schmidt Sciences Trustworthy AI Award.